

HICA-S, Track T4: Local Pruning of Bid-Independent Route Pools

Main results with complete proofs (reader-friendly version)

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How to read this note

The procurement mechanism of HICA-S generates a route pool for every driver *before* any bid is observed, and then runs exact VCG on that pool. This note asks how far such a pool can be reduced, before bidding, by a rule that looks only at the driver whose routes are being deleted.

[Section 1](#) is self-contained. It lists the notation ([Table 1](#)), collects the elementary facts that are used repeatedly ([Section 1.2](#)), and then states and proves the two main results. In plain words:

- **Theorem 1 (local pruning frontier).** A route r of driver i can be deleted safely, using only driver i 's own data, exactly when for every admissible report a *local substitute* is at least as cheap: a route of the same driver that serves part of the orders of r , with the remaining orders sent to the fixed fleet (FD). Deleting all such routes changes no optimal value and no VCG payment, and every route that survives is needed in some instance, so no rule of this type can delete more.
- **Theorem 2 (price of locality).** The surviving pool can still contain one route for each of the $\Theta(n^B)$ bundles of at most B orders, even in instances in which none of these routes is ever used.

[Section 2.7](#) adds two results behind the final version of Algorithm A: the frontier can be computed inside the generation loop ([Proposition 2](#)), and labels that can no longer be completed can be discarded without changing the pool ([Proposition 3](#)), which makes Algorithm A 1.9–3.9× faster on our instances. [Section 2.6](#) explains why pruning partial routes by their cost, in the spirit of rollback pruning, is not part of the final algorithm.

[Section 2](#) contains remarks, the procedures that compute these objects, the NP-hardness of exact non-local pruning, and the computational evidence. Nothing in [Section 1](#) depends on [Section 2](#). [Figure 1](#) shows how the results depend on each other.

1 Main results

1.1 Setting and notation

Orders and the fixed fleet. O is a finite set of orders. Every order $o \in O$ can always be delegated to the fixed-capacity fleet (FD) at a public price $q_o \geq 0$; for $A \subseteq O$ we write $q(A) := \sum_{o \in A} q_o$. The platform publishes a bundle cap $B \geq 1$.

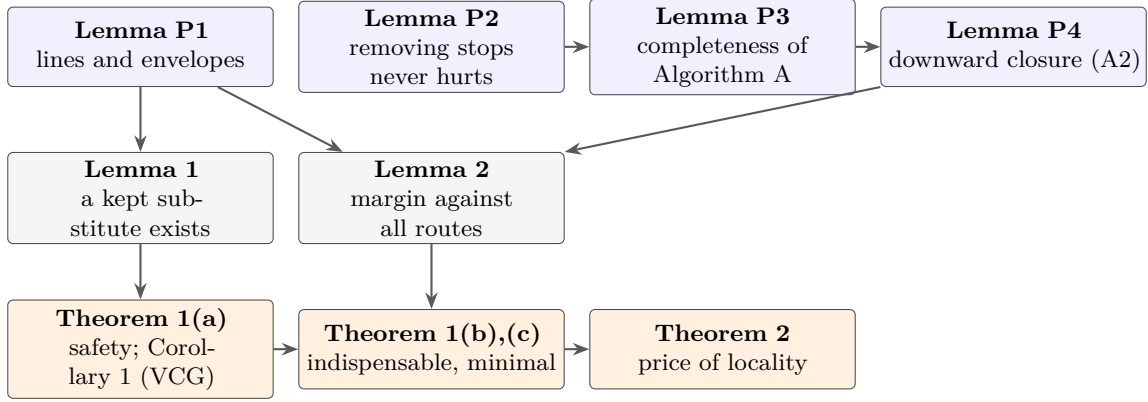


Figure 1: Roadmap. Blue boxes are preliminaries (Section 1.2), white boxes are the lemmas behind the main results, orange boxes are the main results. Lemma P3 is also used by Propositions 2 and 3 in Section 2.7.

Drivers, routes and costs. N is the set of strategic drivers (gigworkers and occasional drivers). Driver i owns a finite route pool R_i , fixed before reports are collected. A route $r \in R_i$ serves a nonempty *bundle* $S_r \subseteq O$ with $|S_r| \leq B$ and has two public nonnegative coefficients, its *pair* $\text{pair}(r) = (K_r, W_r)$: K_r is a distance cost and W_r is working time in hours. If the driver reports the value of time $b \geq 0$, the reported cost of r is

$$c_r(b) = K_r + b W_r. \quad (1)$$

Thus every route is a *line* in the report b , with intercept K_r and slope W_r . The idle option of driver i is the empty route $\mathbf{0}_i$ with $S_{\mathbf{0}_i} = \emptyset$ and $K = W = 0$. We write $\text{pair}(r) \leq \text{pair}(r')$ if $K_r \leq K_{r'}$ and $W_r \leq W_{r'}$; then $c_r(b) \leq c_{r'}(b)$ for every $b \geq 0$.

Reports and the pruning domain. Pruning is required to be correct for all report profiles in a domain $\Theta = [\underline{\theta}, \bar{\theta}]$ with $0 \leq \underline{\theta} < \bar{\theta}$, announced before bidding (in the experiments $\Theta = [18, 25]$ money units per hour).

Winner determination and payments. An *allocation* $\rho = (\rho_j)_{j \in N}$ assigns to every driver one route $\rho_j \in R_j \cup \{\mathbf{0}_j\}$ such that the bundles are pairwise disjoint; orders that no selected route serves are sent to FD. For a report profile $b = (b_j)_{j \in N}$ its cost is

$$C(\rho; b) = \sum_{j \in N} c_{\rho_j}(b_j) + q\left(O \setminus \bigcup_{j \in N} S_{\rho_j}\right). \quad (2)$$

$V(R; b)$ is the minimum of $C(\rho; b)$ over all allocations; this is the set-partitioning winner-determination problem (Algorithm B), in which every order is served exactly once, by a route or by FD, and every driver receives at most one route. $V_{-k}(R; b)$ is the same minimum with driver k removed (Algorithm C). A winner k with route ρ_k^* is paid the Clarke-pivot payment

$$p_k = c_{\rho_k^*}(b_k) + V_{-k}(R; b) - V(R; b). \quad (3)$$

For sub-pools $R'_j \subseteq R_j$ we write $V(R'; b)$ for the same problem on the smaller pools.

Routes as stop sequences. Only the proofs of Lemmas P2 to P4 and of Theorem 2 look inside a route. There, a route of driver i is a sequence of stops (pickup p_o and delivery d_o of each served

order) starting from the point v_0 at the fixed departure time t_0 . At its k -th stop v_k the driver starts service and becomes ready to leave at

$$a_k = \max\{e_{v_k}, r_{k-1} + \tau(v_{k-1}, v_k)\}, \quad r_k = a_k + s_{v_k}, \quad r_0 = t_0, \quad (4)$$

where τ is travel time (minutes), s_v the service time and e_v the opening time of stop v (the release time for a pickup; $e_v = -\infty$ if the stop has no opening time). The max in (4) is the waiting time: a driver who arrives before e_v waits.

Standing assumptions. The following conditions hold throughout.

- (S1) Distances $d(\cdot, \cdot)$ and travel times $\tau(\cdot, \cdot)$ are nonnegative and satisfy the triangle inequality; service times are nonnegative.
- (S2) Opening and release times are lower bounds, met by waiting. Closing times, the end of availability, the capacity, the bundle cap B and (for an occasional driver) the detour budget are upper bounds.
- (S3) For a gigworker with distance coefficient κ , $K = \kappa \cdot (\text{route length})$ and $W = (\text{completion time} - t_0)/60$. For an occasional driver the route ends at her personal destination and $K = \kappa \cdot (\text{route length} - \text{direct length})$, $W = (\text{arrival time} - t_0 - \text{direct time})/60$; both are nonnegative by (S1).

Table 1 lists all symbols used in Section 1.

1.2 Preliminaries

Lines and lower envelopes. Because every cost (1) is a line in the report, the whole analysis reduces to comparing lines on an interval. The following facts are elementary and are used without further comment.

Lemma P1 (Lines and lower envelopes). *Let $g_1, \dots, g_m$ be lines in b and $E(b) = \min_j g_j(b)$ their lower envelope (Fig. 3a).*

- (i) *E is continuous, concave and piecewise linear. Its breakpoints are the values of b at which the lowest line changes.*
- (ii) *For any further line c , the difference $E - c$ is concave and piecewise linear with the same breakpoints. On an interval $[\underline{\theta}, \bar{\theta}]$ its maximum is attained at $\underline{\theta}$, at $\bar{\theta}$ or at a breakpoint in between.*
- (iii) *Two different lines agree at no more than one point. Hence a line that is nonnegative on an interval of positive length and is not identically zero is strictly positive on that interval except at, at most, one point.*
- (iv) *If finitely many strict inequalities between lines hold at b , they also hold at every b' close enough to b .*

Proof. A minimum of affine functions is concave; between two consecutive breakpoints $E - c$ is affine, so its maximum over each piece is at an end of the piece. The difference of two different lines is a nonzero affine function and has at most one root. Item (iv) is continuity. $\square$

Table 1: Notation.

<i>Data</i>	
$O; o, j, x$	set of orders; generic orders
$q_o, q(A)$	FD price of order o ; $q(A) = \sum_{o \in A} q_o$
B	bundle cap: a route serves at most B orders
N	set of strategic drivers (gigworkers and occasional drivers)
$\Theta = [\underline{\theta}, \bar{\theta}]$	pruning domain of reports, announced before bidding, $0 \leq \underline{\theta} < \bar{\theta}$
$b_i; b = (b_i)_{i \in N}$	report of driver i (value of time, per hour); report profile
<i>Routes and costs</i>	
R_i	route pool of driver i , fixed before reports
$\mathbf{0}_i$	idle route of driver i : no orders, $K = W = 0$
S_r	bundle of route r (set of orders it serves), $1 \leq S_r \leq B$
$\text{pair}(r) = (K_r, W_r)$	distance cost and working time (hours) of r
$c_r(b) = K_r + bW_r$	reported cost of r ; a line in b with intercept K_r , slope W_r
$\text{pair}(r) \leq \text{pair}(r')$	$K_r \leq K_{r'}$ and $W_r \leq W_{r'}$; implies $c_r(b) \leq c_{r'}(b)$ for all $b \geq 0$
<i>Allocation and payments</i>	
$\rho = (\rho_j)_{j \in N}$	allocation: one route or $\mathbf{0}_j$ per driver, disjoint bundles, the rest to FD
$C(\rho; b)$	cost (2) of allocation ρ at reports b
$V(R; b), V_{-k}(R; b)$	optimal value; optimal value without driver k
p_k	Clarke-pivot payment (3)
<i>Local frontier (Section 1.3)</i>	
$L_i = (O, q, B, \Theta, R_i)$	local view of driver i : everything a local rule may read
$D(r)$	local substitutes of r : other routes of the same driver serving a subset of S_r , and $\mathbf{0}_i$
$g_{r'}(b)$	substitute line $c_{r'}(b) + q(S_r \setminus S_{r'})$: use r' , send the rest of S_r to FD
$E_r(b)$	FD-completion envelope: the lowest substitute line at b
$\mu_r(b) = E_r(b) - c_r(b)$	local margin of r : how much r saves over its best local substitute
$\mathcal{K}_i^*$	local frontier of driver i : routes with $\mu_r(b) > 0$ for some $b \in \Theta$
Π^*	frontier rule: delete $R_i \setminus \mathcal{K}_i^*$ for every driver
<i>Routing (Lemmas P2–P4, Theorem 2)</i>	
p_o, d_o	pickup and delivery stop of order o
v_0, t_0	start point and (fixed) departure time of a driver
$d(u, v), \tau(u, v)$	distance (km) and travel time (min) from u to v
s_v, e_v, l_v	service time, opening time and closing time of stop v
a_k, r_k	service start and ready time at the k -th stop, (4)
κ	distance coefficient (money per km)
(v, IV, C, t, K, W)	label of Algorithm A: a partial route (stop, orders on board, orders delivered, clock, costs)

Removing stops. The next lemma is the only place where routing enters the theory. Shortening a route obviously shortens its length; that it also never delays a later stop is less obvious, because a driver who arrives earlier may simply wait longer.

Lemma P2 (Removing stops never hurts). *Let σ' be obtained from a stop sequence σ by deleting some stops, both starting at (v_0, t_0) . Under (S1)–(S2):*

- (i) σ' is no longer than σ ;
- (ii) every stop kept in σ' starts service in σ' no later than in σ ;
- (iii) the completion time (for an occasional driver, the arrival time at the destination) of σ' is no later than that of σ .

Neither sequence needs to be feasible.

Proof. (i) is the triangle inequality, applied to every stretch of deleted stops. (ii) Let u and v be consecutive kept stops. In σ the driver travels from u to v through deleted stops, so, since waiting and service are nonnegative and by the triangle inequality, $a_\sigma(v) \geq a_\sigma(u) + s_u + \tau(u, v)$; moreover $a_\sigma(v) \geq e_v$. In σ' , by (4), $a_{\sigma'}(v) = \max\{e_v, a_{\sigma'}(u) + s_u + \tau(u, v)\}$. Hence $a_{\sigma'}(u) \leq a_\sigma(u)$ implies $a_{\sigma'}(v) \leq a_\sigma(v)$. The first kept stop is treated in the same way with $u = v_0$, and induction gives (ii). (iii) follows from (ii) applied to the last stop, or to the destination regarded as a final stop without opening time. $\square$

Algorithm A. Algorithm A generates the pool of each driver by forward label setting. A *label* is a partial route, stored as (v, IV, C, t, K, W) : current stop, orders on board, orders delivered, clock, and accumulated costs. Labels are extended one stop at a time and pass two filters (the final version, [Procedure 2](#) in [Section 2.7](#), processes the batches as tiers of equal event count and also computes the frontier inside the loop):

- *Layer 1 (label dominance).* Labels are processed in *batches*. Within a batch, a label is removed if another label with the same key (v, IV, C) has no later clock and no larger K and W , one of them strictly smaller. Every label that is created is filtered in exactly one batch, and its extensions are filtered in later batches.
- *Layer 2 (per-bundle Pareto).* Among completed routes with the same bundle, only the Pareto-minimal pairs (K, W) are kept, one route per pair.

The returned pool R_i therefore contains no two routes with the same bundle and the same pair.

Lemma P3 (Completeness of Algorithm A). *Let ρ be any feasible route of driver i and R_i the pool returned by Algorithm A. Then R_i contains a route with bundle S_ρ and pair at most $\text{pair}(\rho)$. Consequently, R_i contains exactly one route for each Pareto-minimal pair of each bundle.*

Proof. Follow the prefixes of ρ (the labels it passes through) during the run, and look at the first prefix that does not survive.

Case 1: every prefix survives. Then ρ is completed, and Layer 2 keeps a route with bundle S_ρ and pair at most $\text{pair}(\rho)$.

Case 2: a prefix L is removed by Layer 1. Dominance is irreflexive and transitive, so in the finite batch of L some label L' that dominates L survives. L' is at the same stop, has the same orders on board and the same orders delivered, and has no later clock and no larger costs. Append the remaining stops of ρ to L' . By (4), starting earlier never makes a later service start later, so every remaining stop is served no later than in ρ and all closing times are met. The added distance is

the same and the completion (or arrival) time is no later, so the result is a feasible route ρ' with $S_{\rho'} = S_{\rho}$ and $\text{pair}(\rho') \leq \text{pair}(\rho)$. All prefixes of ρ' up to L' survive. Repeat the argument with ρ' in place of ρ : its first non-surviving prefix lies in a strictly later batch. As there are finitely many batches, Case 1 is eventually reached for some route with bundle S_{ρ} whose pair is at most $\text{pair}(\rho)$, which proves the first claim. A route kept by Layer 2 cannot be dominated by a feasible route ρ'' of the same bundle, because the first claim for ρ'' would produce a kept route dominating it. Conversely, for a Pareto-minimal pair, the first claim produces a kept route with exactly that pair. $\square$

Downward closure. The next property says that taking orders out of a route never makes it more expensive (Fig. 2). It is assumption A2 below, and it holds automatically for the pools of Algorithm A.

Lemma P4 (Pools of Algorithm A are downward closed). *Let R_i be produced by Algorithm A. For every $r' \in R_i$ and every $T \subseteq S_{r'}$ there is $\hat{r} \in R_i \cup \{\mathbf{0}_i\}$ with $S_{\hat{r}} = T$ and $\text{pair}(\hat{r}) \leq \text{pair}(r')$.*

Proof. For $T = \emptyset$ take $\hat{r} = \mathbf{0}_i$. Otherwise delete from r' the pickup and delivery of every order outside T , and call the result σ . By Lemma P2, σ is no longer than r' and serves every remaining stop no later; hence closing times, availability and (for an occasional driver) the detour budget are respected. The load at each remaining stop is that of the orders of T on board, which is at most the original load; precedence is inherited, and $|T| \leq B$. So σ is feasible, and $\text{pair}(\sigma) \leq \text{pair}(r')$ because both its length and its completion (or arrival) time are no larger. Lemma P3 gives a route $\hat{r} \in R_i$ with bundle T and $\text{pair}(\hat{r}) \leq \text{pair}(\sigma) \leq \text{pair}(r')$. $\square$

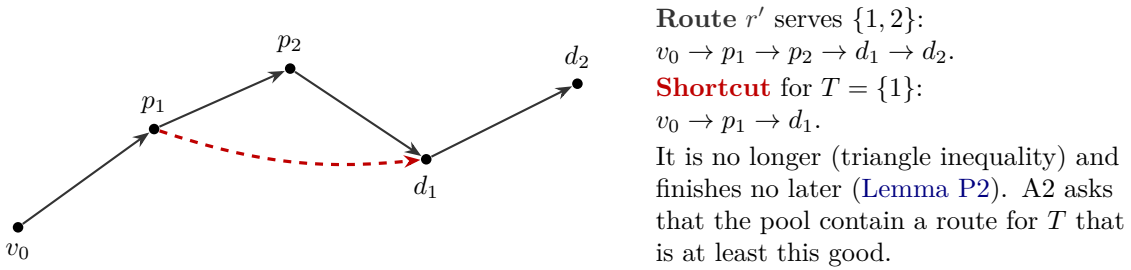


Figure 2: Downward closure (Assumption A2, Lemma P4): deleting the stops of some orders from a feasible route gives a feasible route for the remaining orders whose distance cost and working time are both no larger.

Truthfulness. The mechanism itself is exact VCG on a range that is fixed before bidding; its incentive properties are classical.

Lemma P5 (Truthfulness; Nisan and Ronen, 2007). *If the pools are fixed before reports are observed, every base and removal problem is solved exactly, ties are broken by a rule that reads no report, and costs have the form (1), then truthful reporting is a dominant strategy and every driver's utility is nonnegative.*

Proof. This is the maximal-in-range argument: the utility of driver i equals $V_{-i}(R; b)$ minus the cost of the selected allocation evaluated at i 's true cost and the others' reports, and the mechanism minimises exactly that cost when i reports truthfully. Utility is nonnegative because $\mathbf{0}_i$ is always available. See Nisan and Ronen (2007). $\square$

1.3 The local frontier

Definition 1 (Local view, local rule, safety). The *local view* of driver i is $L_i = (O, q, B, \Theta, R_i)$. A *local pruning rule* Π is a deterministic map that assigns to every local view a set $\Pi(L_i) \subseteq R_i$ of routes to delete; applied to every driver it yields the pools $R_j^\Pi = R_j \setminus \Pi(L_j)$. An *instance* is any finite set of drivers whose pools consist of routes with bundles of size at most B and nonnegative coefficients, in which no two routes of the same driver share both the bundle and the pair. The rule is *safe* if $V(R^\Pi; b) = V(R; b)$ for every instance and every $b \in \Theta^N$.

A local rule reads no report, so the pruned pools are fixed before bidding and [Lemma P5](#) still applies. The ban on duplicates is harmless: duplicates can be merged before pruning without changing any optimal value, and Algorithm A never produces them.

Assumptions on the pool of the driver under study. Theorem 1(b)–(c) and Theorem 2 use the following three properties of a single pool R_i .

Assumption A1 (No duplicates). No two routes of R_i have the same bundle and the same pair.

Assumption A2 (Downward closure: dropping orders never costs more). For every route $r' \in R_i$ and every set $T \subseteq S_{r'}$ of its orders, $R_i \cup \{\mathbf{0}_i\}$ contains a route $\hat{r}$ with $S_{\hat{r}} = T$ and $\text{pair}(\hat{r}) \leq \text{pair}(r')$.

Assumption A3 (No free extra orders). If $r, r' \in R_i$ and $S_r \subsetneq S_{r'}$, then $\text{pair}(r) \neq \text{pair}(r')$: serving additional orders costs some distance or some time.

[Assumption A1](#) holds in every instance by [Definition 1](#); [Assumptions A1](#) and [A2](#) hold for every pool of Algorithm A by [Lemmas P3](#) and [P4](#) (see [Fig. 2](#)). [Assumption A3](#) only excludes serving extra orders at exactly zero extra distance and zero extra time.

The envelope, the margin and the frontier. When the platform considers deleting a route r of driver i without looking at other drivers, the only replacements it can trust are those that involve nobody else: another route r' of driver i that serves *part* of S_r , with the remaining orders $S_r \setminus S_{r'}$ sent to FD at their public price. At report b such a replacement costs $g_{r'}(b) = c_{r'}(b) + q(S_r \setminus S_{r'})$, again a line in b .

Definition 2 (Substitutes, envelope, margin, frontier). For $r \in R_i$, the set of *local substitutes* is

$$D(r) = \{ r' \in R_i \cup \{\mathbf{0}_i\} : r' \neq r, S_{r'} \subseteq S_r \}.$$

The *FD-completion envelope* and the *local margin* of r are

$$E_r(b) = \min_{r' \in D(r)} \{ c_{r'}(b) + q(S_r \setminus S_{r'}) \}, \quad \mu_r(b) = E_r(b) - c_r(b).$$

The *local frontier* of driver i is $\mathcal{K}_i^* = \{ r \in R_i : \mu_r(b) > 0 \text{ for some } b \in \Theta \}$, and the *frontier rule* Π^* deletes $R_i \setminus \mathcal{K}_i^*$ for every driver.

In words, $\mu_r(b)$ is how much route r saves, at report b , compared with the best way of doing without r that uses only driver i and FD. A route belongs to the frontier if it is strictly the best local option for at least one admissible report. Since $\mathbf{0}_i \in D(r)$, always $E_r(b) \leq q(S_r)$. [Figure 3](#) illustrates the definition. By [Lemma P1\(ii\)](#), membership in $\mathcal{K}_i^*$ is decided by evaluating μ_r at $\underline{\theta}$, at $\bar{\theta}$ and at the breakpoints of E_r in between ([Procedure 1](#)).

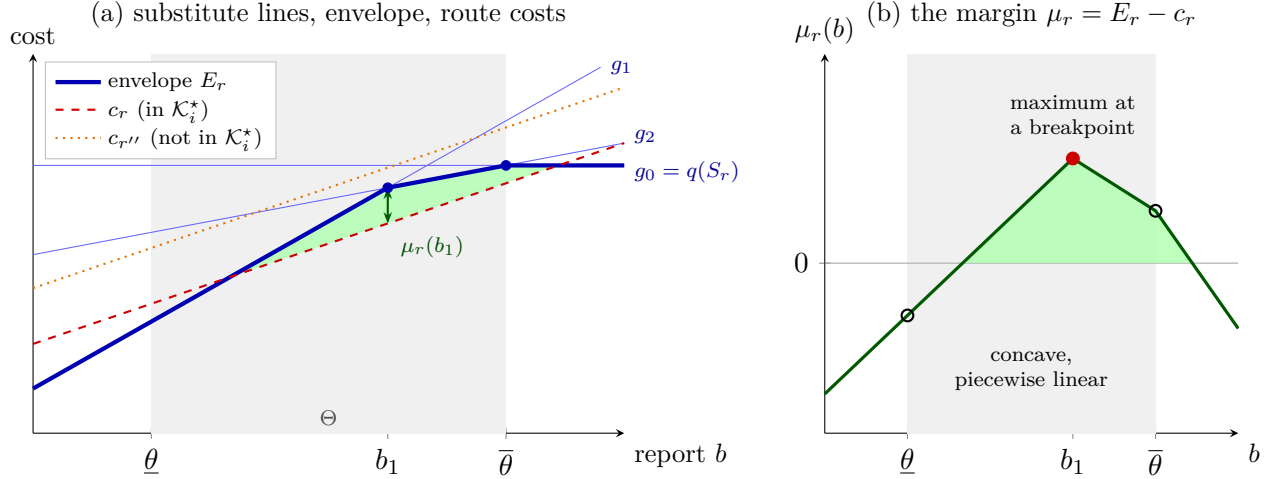


Figure 3: The envelope and the margin. (a) Each thin line is a local substitute $g_{r'}(b) = c_{r'}(b) + q(S_r \setminus S_{r'})$; g_0 is the idle option (all of S_r to FD). The thick line is their lower envelope E_r , with breakpoints (dots) where the lowest substitute changes. Route r (dashed) lies below E_r on the green region, so it is in the frontier. Route r'' (dotted) lies above E_r everywhere, so some local substitute is always at least as cheap and r'' can be deleted. (b) The margin $\mu_r = E_r - c_r$ is concave and piecewise linear, so its maximum over Θ is attained at $\underline{\theta}$, at $\bar{\theta}$ or at a breakpoint (Lemma P1); here at the breakpoint b_1 .

Example 1 (Twin instances). A gigworker g faces two orders with $q_{o_1} = 15$, $q_{o_2} = 45$, and $\Theta = [18, 25]$. Her pool has three routes:

$$r_a : \{o_1\}, (6, 0.6); \quad r_b : \{o_2\}, (8, 0.8); \quad r_c : \{o_1, o_2\}, (11, 1.2).$$

Frontier. r_a : its only substitute is $\mathbf{0}_g$, so $E_{r_a} = q_{o_1} = 15 < 6 + 0.6b$ on Θ ; $r_a \notin \mathcal{K}_g^*$ (FD is always cheaper). r_b : $E_{r_b} = 45 > 8 + 0.8b$ on Θ , so $r_b \in \mathcal{K}_g^*$. r_c : at $b = 18$ the three substitutes cost $6 + 10.8 + 45 = 61.8$ (r_a, o_2 to FD), $8 + 14.4 + 15 = 37.4$ (r_b, o_1 to FD) and 60 (idle), so $E_{r_c}(18) = 37.4 > 32.6 = c_{r_c}(18)$ and $r_c \in \mathcal{K}_g^*$.

Instance I_0 (g alone). r_c costs less than r_b plus FD for o_1 iff $11 + 1.2b < 23 + 0.8b$, i.e. $b < 30$; so r_c is optimal on all of Θ and r_b is never used.

Instance I_1 (I_0 plus a driver who serves o_1 at cost 0.5). At $b = 18$ the optimum is $22.4 + 0.5 = 22.9$ and uses r_b ; without r_b it is 32.6.

A rule that deletes r_b — for instance a hull across bundles, under which r_a dominates r_b — is right on I_0 and wrong on I_1 . Driver g 's data cannot tell the two instances apart. This is Theorem 1(b) in its smallest form.

1.4 Theorem 1: the local pruning frontier

Theorem 1 (Local pruning frontier). Let $\underline{\theta} < \bar{\theta}$.

- (a) **Safety.** Π^* is safe. Moreover $V_{-k}(R^{\Pi^*}; b) = V_{-k}(R; b)$ for every driver k and every $b \in \Theta^N$.
- (b) **Indispensability.** Let R_i satisfy Assumptions A2 and A3 and $r \in \mathcal{K}_i^*$. There are an instance that contains driver i with the same local view L_i , a report profile $b \in \Theta^N$ and a number $\gamma > 0$ such that every optimal allocation assigns r to i and $V(R \setminus \{r\}; b) \geq V(R; b) + \gamma/2$.
- (c) **Minimality.** Every safe local rule, applied to a local view that satisfies Assumptions A2 and A3, deletes only routes outside $\mathcal{K}_i^*$. Hence $\mathcal{K}_i^*$ is the unique smallest pool that a safe local rule can retain, and Π^* attains it.

The proof of (a) rests on [Lemma 1](#): a deleted route can always be replaced by a route that is itself kept, not merely by some route that might have been deleted too. The proof of (b) rests on [Lemma 2](#), which extends the margin from substitutes (routes serving a subset of S_r) to *all* other routes of the driver.

Lemma 1 (A kept substitute always exists). *Let $\underline{\theta} < \bar{\theta}$. For every $r \in R_i \setminus \mathcal{K}_i^*$ and every $b \in \Theta$ there is $u \in \mathcal{K}_i^* \cup \{\mathbf{0}_i\}$ with $S_u \subseteq S_r$ and*

$$c_u(b) + q(S_r \setminus S_u) \leq c_r(b).$$

Proof. Fix b . Call every route $\rho \in R_i \cup \{\mathbf{0}_i\}$ with $S_\rho \subseteq S_r$ an *option* for S_r (this includes r itself), and let $\Phi(\rho) = c_\rho(b) + q(S_r \setminus S_\rho)$ be its cost at b : driver i serves S_ρ by ρ and FD serves the rest of S_r . Note $\Phi(r) = c_r(b)$.

Step 1: choose a cheapest option with a smallest bundle. Let m be the smallest value of Φ ; among the options with $\Phi = m$ take one with the fewest orders, and let T be its bundle and $k = |T|$. If $k = 0$, this option is $\mathbf{0}_i$ and $u = \mathbf{0}_i$ satisfies the claim, because $\Phi(\mathbf{0}_i) = m \leq \Phi(r) = c_r(b)$.

Step 2: a strict comparison at b . Let $k \geq 1$ and let M_T be the set of routes with bundle T and $\Phi = m$, i.e. with the same cost at b as the chosen option. We claim that every $\rho \in M_T$ is strictly cheaper at b than each of its own substitutes ρ'' that is not in M_T :

$$c_{\rho''}(b) + q(T \setminus S_{\rho''}) > c_\rho(b). \quad (5)$$

If $S_{\rho''} \subsetneq T$, then ρ'' is an option for S_r with fewer than k orders, so $\Phi(\rho'') > m = \Phi(\rho)$ by the choice of k ; removing the common term $q(S_r \setminus T)$ from both sides gives (5). If $S_{\rho''} = T$, then $\Phi(\rho'') \geq m$ means $c_{\rho''}(b) \geq c_\rho(b)$, and equality would put ρ'' in M_T .

Step 3: break ties by moving the report slightly ([Fig. 4](#)). The routes of M_T have the same bundle and the same cost at b . By [Assumption A1](#) their pairs differ, so their slopes W differ. If $b < \bar{\theta}$, let $u \in M_T$ be the route with the smallest slope. For $b' > b$ close to b , u is strictly cheaper than every other route of M_T , and the finitely many strict inequalities (5) still hold at b' ([Lemma P1\(iv\)](#)). So at b' route u is strictly cheaper than every one of its substitutes, that is $\mu_u(b') > 0$, and $u \in \mathcal{K}_i^*$. If $b = \bar{\theta}$, take the route with the largest slope and $b' < b$ instead.

Step 4: conclusion. $S_u = T \subseteq S_r$ and $c_u(b) + q(S_r \setminus T) = \Phi(u) = m \leq \Phi(r) = c_r(b)$. (Since $u \in \mathcal{K}_i^*$ and $r \notin \mathcal{K}_i^*$, automatically $u \neq r$.) $\square$

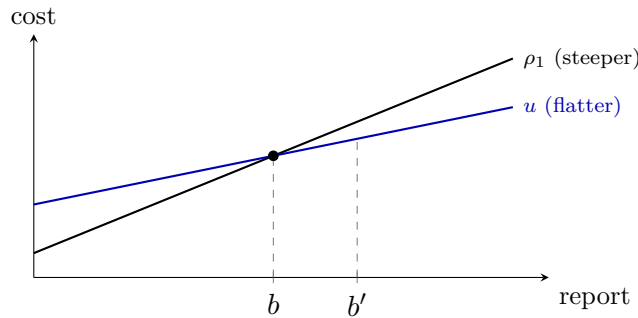


Figure 4: Tie-breaking in Step 3 of [Lemma 1](#). Two routes with the same bundle cost the same at b . Just to the right of b the flatter one (the one with less working time) is strictly cheaper; just to the left, the steeper one. This is why ties cannot block membership in the frontier.

Proof of Theorem 1(a). Fix an instance, a profile $b \in \Theta^N$ and an optimal allocation ρ for the full pools. For every driver j whose route ρ_j is deleted ($\rho_j \notin \mathcal{K}_j^* \cup \{\mathbf{0}_j\}$), [Lemma 1](#) gives a kept route $u_j \in \mathcal{K}_j^* \cup \{\mathbf{0}_j\}$ with $S_{u_j} \subseteq S_{\rho_j}$ and $c_{u_j}(b_j) + q(S_{\rho_j} \setminus S_{u_j}) \leq c_{\rho_j}(b_j)$. Give u_j to driver j and send $S_{\rho_j} \setminus S_{u_j}$ to FD. Bundles only shrink, so they stay disjoint; every order is still served exactly once; and no other driver is affected, so all replacements can be made at the same time. The new allocation uses only kept routes and costs at most $V(R; b)$. Hence $V(R^{\Pi^*}; b) \leq V(R; b)$; the reverse inequality holds because the pruned pools are smaller. Removing a driver k gives another instance in which every remaining driver has the same local view, so the same argument applies to V_{-k} . $\square$

Corollary 1 (VCG outcomes are preserved). *Let $\mathcal{K}^* = \bigcup_{i \in N} \mathcal{K}_i^*$ and $b \in \Theta^N$.*

- (i) *Running Algorithms B and C on the pruned pools is VCG on a range fixed before bidding, hence truthful and individually rational ([Lemma P5](#)).*
- (ii) *Suppose the full-pool mechanism breaks ties among optimal allocations first by using as few routes outside $\mathcal{K}^*$ as possible, and then by the same report-independent rule as the pruned mechanism. Then both select the same allocation, and every payment (3) coincides.*
- (iii) *If the full problem has a unique optimal allocation at b , that allocation uses only routes of $\mathcal{K}^*$, and (ii) holds under any tie-breaking rule.*

Proof. (i) is [Lemma P5](#). (ii) The proof of Theorem 1(a) turns every optimal allocation into one of no larger cost that uses only routes of $\mathcal{K}^*$. So the optimal allocations that use no route outside $\mathcal{K}^*$ are exactly the optimal allocations of the pruned problem, which has the same optimal value; the remaining tie-breaking criteria are the same in both mechanisms. V and every V_{-k} coincide by Theorem 1(a), so (3) coincides. The first tie-breaking criterion reads no report, so [Lemma P5](#) also applies to the full-pool mechanism. (iii) If the unique optimum used a route outside $\mathcal{K}^*$, the replacement above would produce a different allocation of no larger cost. $\square$

Lemma 2 (Margin against all other routes). *Let R_i satisfy [Assumptions A2](#) and [A3](#), $\underline{\theta} < \bar{\theta}$, and $r \in \mathcal{K}_i^*$ with $S := S_r$. There are $b^* \in \Theta$ and $\gamma > 0$ such that*

$$c_{r'}(b^*) + q(S \setminus S_{r'}) \geq c_r(b^*) + \gamma \quad \text{for every } r' \in R_i \cup \{\mathbf{0}_i\}, r' \neq r. \quad (6)$$

In words: at report b^* , replacing r by any other route of driver i — whatever orders that route serves — and sending the orders of S that it misses to FD costs at least γ more. Orders outside S that r' happens to serve are not counted on the left; they are dealt with in the proof of Theorem 1(b).

Proof. Step 1: an interval of positive margin. Since $r \in \mathcal{K}_i^*$, $\mu_r(\bar{b}) > 0$ for some $\bar{b} \in \Theta$. The margin is continuous ([Lemma P1](#)), so $\mu_r > 0$ on a whole interval $J \subseteq \Theta$ of positive length around $\bar{b}$; if $\bar{b}$ is an end point of Θ , J lies on the inner side, which is possible because $\underline{\theta} < \bar{\theta}$.

Step 2: three cases for r' ([Fig. 5](#)). Let $b \in J$.

- (i) $S_{r'} \subseteq S$. Then r' is a substitute of r , so the left-hand side of (6) is at least $E_r(b) = c_r(b) + \mu_r(b)$.
- (ii) $S_{r'}$ overlaps S partially, i.e. $T := S \cap S_{r'} \neq S$ and $S_{r'} \not\subseteq S$. By [Assumption A2](#) applied to r' and T , there is a route $\hat{r}$ with $S_{\hat{r}} = T$ and $c_{\hat{r}}(b) \leq c_{r'}(b)$. As $T \neq S$, $\hat{r}$ is a substitute of r . Since $S \setminus S_{r'} = S \setminus T$,

$$c_{r'}(b) + q(S \setminus S_{r'}) \geq c_{\hat{r}}(b) + q(S \setminus T) \geq E_r(b) = c_r(b) + \mu_r(b).$$

- (iii) $S \subsetneq S_{r'}$. Then $q(S \setminus S_{r'}) = 0$. By [Assumption A2](#) with $T = S$ there is $\hat{r}$ with $S_{\hat{r}} = S$ and $c_{\hat{r}}(b) \leq c_{r'}(b)$. If $\hat{r} = r$, then $c_{r'}(b) \geq c_r(b)$. If $\hat{r} \neq r$, then $\hat{r}$ is a substitute of r with the same bundle, so $c_{\hat{r}}(b) \geq E_r(b) > c_r(b)$. In both cases the line $h_{r'}(b) := c_{r'}(b) - c_r(b)$ is nonnegative on J . By

Assumption A3, $\text{pair}(r') \neq \text{pair}(r)$, so $h_{r'}$ is not identically zero, and by **Lemma P1**(iii) it is strictly positive on J except at, at most, one point.

Step 3: choice of b^ and γ .* Pick $b^* \in J$ different from the finitely many exceptional points of case (iii); this is possible because J has positive length. Put $\gamma = \min\{\mu_r(b^*), \min_{r' \text{ in case (iii)}} h_{r'}(b^*)\} > 0$. Cases (i) and (ii) give (6) because $\mu_r(b^*) \geq \gamma$, and case (iii) because $h_{r'}(b^*) \geq \gamma$. $\square$

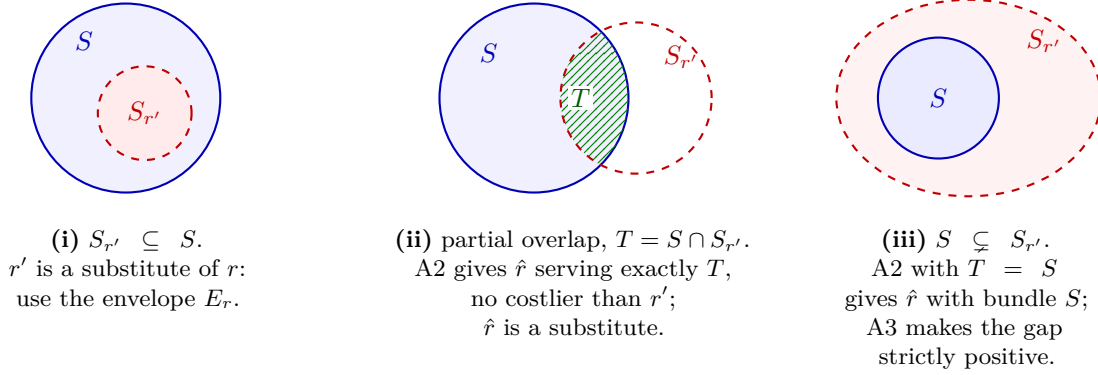


Figure 5: The three ways another route r' of the same driver can relate to the bundle $S = S_r$ (proof of **Lemma 2**). In (ii), downward closure replaces r' by a route $\hat{r}$ that serves only the common orders T and is no more expensive; $\hat{r}$ is then a substitute of r , so the envelope applies.

Proof of Theorem 1(b). Let b^* and γ be as in **Lemma 2** and fix $0 < \varepsilon \leq \gamma/(2B)$.

Construction (Fig. 6). The instance consists of driver i with report b^* and, for every order $x \notin S$, a helper driver j_x whose pool is a single route serving only $\{x\}$ with $K = \varepsilon$ and $W = 0$ (so its report does not matter). The data O, q, B, Θ, R_i are unchanged, so driver i has local view L_i .

Evaluation. Fix the route $r' \in R_i \cup \{\mathbf{0}_i\}$ that driver i receives, and complete the allocation as cheaply as possible:

- orders of $S_{r'}$ are served by i ;
- orders of $S \setminus S_{r'}$ can only go to FD, since no helper serves them;
- every order $x \notin S \cup S_{r'}$ is served by its helper or by FD, at cost $e_x := \min\{\varepsilon, q_x\} \leq \varepsilon$;
- for orders $x \in S_{r'} \setminus S$ the helper stays idle.

The cheapest allocation in which i receives r' therefore costs

$$F(r') = c_{r'}(b^*) + q(S \setminus S_{r'}) + \sum_{x \notin S \cup S_{r'}} e_x, \quad \text{in particular} \quad F(r) = c_r(b^*) + \sum_{x \notin S} e_x.$$

Subtracting, and using (6) and $|S_{r'} \setminus S| \leq B$,

$$F(r') - F(r) = \underbrace{c_{r'}(b^*) + q(S \setminus S_{r'}) - c_r(b^*)}_{\geq \gamma} - \underbrace{\sum_{x \in S_{r'} \setminus S} e_x}_{\leq B\varepsilon \leq \gamma/2} \geq \frac{\gamma}{2} \quad \text{for every } r' \neq r.$$

Hence every optimal allocation gives r to driver i , and $V(R \setminus \{r\}; b) = \min_{r' \neq r} F(r') \geq F(r) + \gamma/2 = V(R; b) + \gamma/2$. $\square$

The helpers make every order outside S almost free. A route r' that serves such orders therefore gains almost nothing from them (at most $B\varepsilon$), while by **Lemma 2** it loses at least γ on the orders of S .

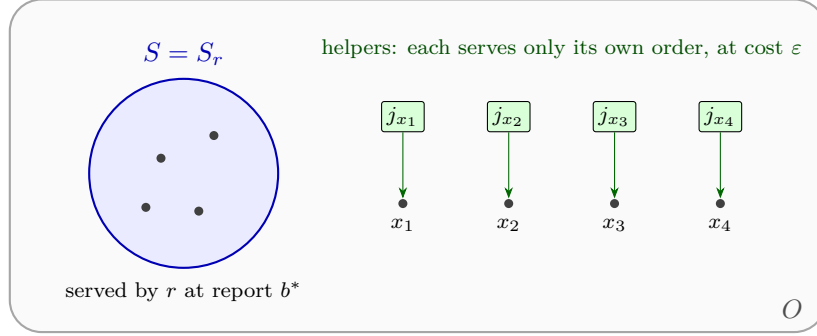


Figure 6: The instance in the proof of Theorem 1(b). Orders outside S become almost free (cost ε), orders inside S can only be served by driver i or by FD. Driver i 's own data are unchanged, so no local rule can distinguish this instance from the original one.

Proof of Theorem 1(c). Suppose a local rule Π deletes some $r \in \mathcal{K}_i^*$ from a local view L_i that satisfies [Assumptions A2](#) and [A3](#). In the instance of part (b), driver i has local view L_i , so Π deletes r there as well (and possibly other routes). Every allocation of the pruned pools is an allocation of $R \setminus \{r\}$, hence $V(R^\Pi; b) \geq V(R \setminus \{r\}; b) > V(R; b)$ and Π is not safe. So every safe local rule deletes only routes outside $\mathcal{K}_i^*$; Π^* deletes all of them and is safe by part (a). $\square$

1.5 Theorem 2: the price of locality

Theorem 1 says what a local rule must keep. Theorem 2 shows that this can be a great deal, and that all of it can be useless.

Theorem 2 (Price of locality). *Fix $B \geq 1$. For every $n \geq B$ there is a Euclidean instance I_n with n orders, one gigworker g and n occasional drivers such that*

- (i) R_g contains exactly one route for each bundle of at most B orders, all of them in $\mathcal{K}_g^*$; hence every safe local rule keeps $\sum_{k=1}^B \binom{n}{k} = \Theta(n^B)$ routes of g ;
- (ii) for every report profile in Θ^{n+1} , no optimal allocation assigns a route to g .

Construction (Fig. 7). All n orders are picked up at the same point P and delivered at the same point D , with $d(P, D) = \ell$, release time 0, non-binding deadlines, service time s (minutes) per stop and a common FD price q_0 . The gigworker starts at A , with $d(A, P) = a$, at time 0, with capacity at least B and distance coefficient $\kappa > 0$. Each occasional driver $\omega_1, \dots, \omega_n$ travels from P to D , departs at time 0 and has detour budget $2s$. Let $T_0 = \tau(A, P) + \tau(P, D)$. The parameters satisfy

$$q_0 > 2s\bar{\theta}/60, \tag{C1}$$

$$\kappa(a + \ell) + \underline{\theta}(T_0 + 2s)/60 < q_0, \tag{C2}$$

$$\kappa(a + \ell) > B(\bar{\theta} - \underline{\theta})2s/60. \tag{C3}$$

For example, $B = 3$, $s = 2$ min, $\Theta = [18, 25]$, $\kappa = 0.5$, $a = 2$ km, $\ell = 6$ km, speed 20 km/h and $q_0 = 20$ give $T_0 = 24$ min and (C1) $20 > 1.67$, (C2) $12.4 < 20$, (C3) $4 > 1.4$. (Condition (C3) depends on B ; with these numbers it holds for $B \leq 8$.)

Proof. Step 1: the gigworker's pool. For a bundle S with $|S| = k \leq B$, every sequence that performs all pickups at P before all deliveries at D has length $a + \ell$ and completion time $T_0 + 2sk$ (no waiting). Every other sequence returns from D to P at least once, so it is at least 2ℓ longer and finishes later;

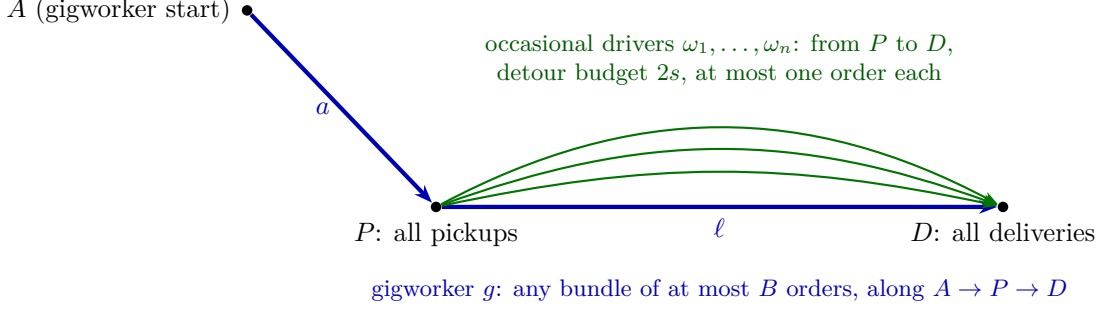


Figure 7: The instance of Theorem 2. The gigworker can serve every bundle of up to B orders along the same path, so her pool has one route per bundle, and none of these routes can be removed by a local rule. Yet the occasional drivers serve every order more cheaply, one each, so the gigworker never wins.

it is Pareto-dominated. Hence $R_g = \{r_S\}$ with $K_{r_S} = \kappa(a + \ell)$ and $W_{r_S} = (T_0 + 2sk)/60$, one route per bundle. [Assumptions A1](#) and [A2](#) hold by [Lemma P4](#), and [Assumption A3](#) holds because W_{r_S} increases strictly with $|S|$.

Step 2: every route is in the frontier. Take $b = \underline{\theta}$ and a bundle S with $|S| = k$. The substitutes of r_S are $r_{S'}$ for $\emptyset \neq S' \subsetneq S$ and $\mathbf{0}_g$. For the former,

$$c_{r_{S'}}(b) + q_0(k - |S'|) - c_{r_S}(b) = (k - |S'|)(q_0 - 2sb/60) > 0$$

by (C1), since the two routes have the same K and differ by $2s(k - |S'|)$ minutes of working time. For the idle option,

$$q_0k - c_{r_S}(b) = (q_0 - c_{r_{\{x\}}}(b)) + (k - 1)(q_0 - 2sb/60) > 0$$

by (C2) and (C1), where $r_{\{x\}}$ is any single-order route. Hence $\mu_{r_S}(\underline{\theta}) > 0$, so $r_S \in \mathcal{K}_g^*$, and by Theorem 1(c) every safe local rule keeps it. This proves (i).

Step 3: the gigworker never wins. Each ω_j can serve any single order with zero extra distance and extra time $2s$ (the two service stops), and cannot serve two orders, which would need an extra time of at least $4s$. Take an allocation in which g serves r_S with $|S| = k$. The $n - k$ orders outside S occupy at most $n - k$ occasional drivers, so at least k of them are idle. Give the orders of S to k idle occasional drivers and make g idle. The reported cost changes by at most

$$-c_{r_S}(b_g) + k\bar{\theta} \frac{2s}{60} \leq -\kappa(a + \ell) - \underline{\theta} \frac{T_0 + 2sk}{60} + k\bar{\theta} \frac{2s}{60} \leq -\kappa(a + \ell) + B(\bar{\theta} - \underline{\theta}) \frac{2s}{60} < 0$$

by (C3). So no allocation that uses a route of g is optimal. This proves (ii). $\square$

The theorem makes the asymmetry between the driver classes precise: nothing bounds the pool that no local rule can shrink except feasibility. A tight detour budget restricts feasibility; time windows alone do not. [Remark 3](#) shows that an occasional driver with a generous detour budget is exposed in exactly the same way.

2 Remarks, algorithms and computational evidence

This section is not needed for the proofs of [Section 1](#). It collects remarks on the scope of the results, the procedures that compute them, and the evidence from the audits and experiments.

2.1 What the frontier is not

Remark 1 (Three ways to beat the frontier). The minimality in Theorem 1(c) is relative to local rules (Definition 1). The frontier can be beaten in three ways, each of which leaves that class. (i) *Non-local rules* that also read other drivers' public data remain bid-independent and keep truthfulness; in the instances of Theorem 2 such a rule could delete every route of the gigworker. Deciding exactly what a non-local rule may delete is, however, NP-hard (Proposition 1). (ii) *Instance-specific rules*, which only need to be correct on the instance at hand, can keep far less: on five experimental instances only 0.4–1.5% of generated routes are optimal for any of 1,000 sampled report profiles, whereas the frontier keeps 3.2–23.1% of the pool. (iii) *Restricting the competitors* to geometrically realisable drivers weakens the safety requirement; Remark 2 bounds the effect.

Proposition 1 (Exact non-local pruning is hard). *For $B \geq 3$ it is NP-hard to decide, given the pools of all drivers, the prices q and a route r , whether $V(R \setminus \{r\}; b) = V(R; b)$ for every $b \in \Theta^N$, even if every route lies in the local frontier of its driver and no cost depends on the report.*

Proof. Reduction from exact cover by 3-sets (Garey and Johnson, 1979): a universe U with $|U| = 3m$ and a family $\mathcal{S}$ of 3-subsets of U . The orders are $U \cup \{o^*\}$, all with $q_o = 1$. For each $F \in \mathcal{S}$ a driver owns one route serving F at cost 0; a driver h owns one route for each single order of $U \cup \{o^*\}$, each at cost 0; and a driver i owns one route r serving $\{o^*\}$ with $K_r = \frac{1}{2}$, $W_r = 0$. Every route is cheaper than the FD price of its bundle and its driver has no route with a smaller bundle, so every route lies in its driver's frontier.

If an exact cover exists, the sets cover U and h serves o^* : $V(R) = V(R \setminus \{r\}) = 0$. Otherwise a maximum packing of $\mathcal{S}$ leaves $u \geq 3$ elements uncovered, and any other packing leaves at least $u + 3$. Without r , h serves one of the $u + 1$ uncovered orders and the rest go to FD, so the optimal value is u . With r , h serves an uncovered element and r serves o^* , giving $u - 1 + \frac{1}{2} = u - \frac{1}{2} < u$. Hence r can be deleted safely if and only if an exact cover exists. $\square$

Remark 2 (Realisable competitors). The helpers j_x in the proof of Theorem 1(b) are abstract. They can be replaced by genuine occasional drivers whose origin and destination are the pickup and delivery of x and whose detour budget equals the total service time s_x at the two stops. Under a general-position condition (no other order has its pickup and delivery, in this order, on the segment from p_x to d_x) such a driver can serve x and nothing else, with zero extra distance and working time $s_x/60$, so at report $\underline{\theta}$ its cost is $\lambda_x = \underline{\theta}s_x/60$ (1.20 in the audit setting with $s_x = 4$ min; 3.00 with the five-minute service of the experiments). Repeating the proof of Theorem 1(b) with $\min\{\lambda_x, q_x\}$ in place of $\min\{\varepsilon, q_x\}$ shows that every frontier route whose margin γ in Lemma 2 exceeds $\sum_{x \in S_{r'} \setminus S} \min\{\lambda_x, q_x\}$ for all r' (for instance $\gamma > B\underline{\theta} \max_x s_x/60$) stays indispensable against such competitors. In a synthetic audit, 556 of 563 frontier routes (98.8%) did. Whether a safe local rule could exploit this credit in general is open: a gigworker route with waiting slack can absorb an extra order at zero extra working time.

2.2 Loose feasibility, not the driver class

Remark 3 (Theorem 2 for occasional drivers). The proof of Theorem 2 uses only that every bundle is feasible, that one sequence per bundle is Pareto-minimal, and that K is constant and W affine in $|S|$. It therefore applies to an occasional driver with origin A , destination D and detour budget at least $\tau(A, P) + \tau(P, D) - \tau(A, D) + 2sB$ in place of g . Her routes have $K_{r_S} = \kappa(a + \ell - d(A, D))$ and $W_{r_S} = (T_0 - \tau(A, D) + 2s|S|)/60$, and the proof holds with $a + \ell$ and T_0 replaced by $a + \ell - d(A, D)$ and $T_0 - \tau(A, D) \geq 0$. With $P = (0, 0)$, $D = (6, 0)$, $A = (0, 4)$ (km), $\kappa = 1$, and B, s, Θ , speed and q_0 as in the example, (C1)–(C3) hold (the left-hand side of (C3) is 2.79 against 1.40, and (C2) reads

6.50 < 20), and a detour budget of 20.4 min suffices, below the 30 min used in the experiments. What Theorem 2 separates is loose from tight feasibility, not one driver class from the other.

Remark 4 (Choice of the evaluation point). The proof of Theorem 2 evaluates the margin at $\underline{\theta}$. Any $b \in \Theta$ works if (C2) is stated at that b ; the manuscript draft of 2026-09-30 uses $\bar{\theta}$ (giving $15.67 < 20$). Using $\underline{\theta}$ gives the weakest condition.

2.3 Monotonicity and the choice of Θ

Remark 5 (Monotonicity). E_r is nondecreasing in every FD price q_o , and a larger Θ can only add reports at which $\mu_r > 0$. For a fixed pool, the frontier therefore grows with the FD tariff and with Θ : local pruning is strong when FD is cheap and weak when it is expensive. This is what the experiments show (Table 4).

Remark 6 (Reports outside Θ). The mechanism does not restrict reports. Because the pruned pools are fixed before bidding, the pruned mechanism is VCG on a fixed range and Lemma P5 applies at every report profile, including profiles outside Θ^N . The domain matters only for *outcome equivalence* with the full pool, which holds at the truthful profile whenever the types lie in Θ . Taking $\Theta = [0, \infty)$ extends the equivalence to all reports at the price of a larger frontier; Definition 2 is written with “for some b ” so that it covers this case.

2.4 Computing the frontier

Remark 7 (Complexity and numerical tolerance). $|D(r)| \leq 2^B f$, where f bounds the number of routes per bundle. By Lemma P1(ii), membership in $\mathcal{K}_i^*$ is decided by evaluating μ_r at the end points of Θ and at the breakpoints of E_r , in $O(|D(r)| \log |D(r)|)$ time; the frontier of a driver costs $O(|R_i| 2^B f \log(2^B f))$. In floating-point arithmetic the test should err on the safe side: delete a route only if the largest margin is below zero by more than a small tolerance. The frontier is a post-processing step: it does not speed up enumeration, but it shrinks every problem solved by Algorithms B and C.

Procedure 1 FRONTIER(R_i, q, Θ): the rule Π^* for one driver

Input: pool R_i with bundles S_r and pairs (K_r, W_r) ; FD prices q ; $\Theta = [\underline{\theta}, \bar{\theta}]$

Output: local frontier $\mathcal{K}_i^* \subseteq R_i$

```

1:  $\mathcal{K}_i^* \leftarrow \emptyset$ 
2: for each  $r \in R_i$  do
3:    $D(r) \leftarrow \{r' \in R_i \cup \{\mathbf{0}_i\} \setminus \{r\} : S_{r'} \subseteq S_r\}$ 
4:   for each  $r' \in D(r)$  form the line  $g_{r'}(b) = K_{r'} + q(S_r \setminus S_{r'}) + b W_{r'}$ 
5:   compute the lower envelope  $E_r = \min_{r' \in D(r)} g_{r'}$  on  $\Theta$   $\triangleright O(|D(r)| \log |D(r)|)$ 
6:    $\mathcal{B} \leftarrow \{\underline{\theta}, \bar{\theta}\} \cup \{\text{breakpoints of } E_r \text{ in } \Theta\}$ 
7:   if  $\max_{b \in \mathcal{B}} (E_r(b) - K_r - b W_r) > 0$  then  $\triangleright \mu_r$  is concave
8:     add  $r$  to  $\mathcal{K}_i^*$ 
9:   end if
10: end for
11: return  $\mathcal{K}_i^*$ 

```

2.5 Relation to LP duality and to the literature

Remark 8 (LP view). In the linear relaxation of the winner-determination problem let π_o be the dual of order o and $\sigma_i \leq 0$ that of driver i 's one-route constraint. The FD columns impose $\pi_o \leq q_o$. If

$S_{r'} \subseteq S_r$ and $c_r \geq c_{r'} + q(S_r \setminus S_{r'})$, then, since both routes belong to driver i , σ_i cancels and

$$\bar{c}_r - \bar{c}_{r'} = c_r - c_{r'} - \pi(S_r \setminus S_{r'}) \geq \sum_{o \in S_r \setminus S_{r'}} (q_o - \pi_o) \geq 0$$

for every dual vector with $\pi \leq q$. So Π^* deletes a route when, at each admissible report, a route of the same driver has no larger reduced cost over the dual region cut by the outside option. Equivalently, r is *redundant* in the sense of Sol (1994), the combination consisting of r' and the FD columns of $S_r \setminus S_{r'}$. The bound $\pi \leq q$ is all a single driver can know about the duals; the construction of Theorem 1(b) realises $\pi \approx q$ on S_r and $\pi \approx 0$ elsewhere. Restricting $D(r)$ to routes of the same bundle and taking $\Theta = [0, \infty)$ turns $\mathcal{K}_i^*$ into the extreme supported points of each bundle's (K, W) frontier (Ehrgott, 2005).

Remark 9 (Literature). Column dominance in set covering (Beasley, 1987), the subsumed-subsets test of Müller (1998), and dominance and redundancy in column generation (Vanderbeck, 1994; Sol, 1994), as summarised by Lübbecke and Desrosiers (2005), all compare columns at known costs or at a fixed dual vector. Three features of Theorem 1 have no counterpart there: costs depend on a report ranging over a domain fixed by the mechanism designer, and the substitute may change with the report; safety is required for the integer optimum and for every Clarke-pivot removal problem; and the converse (Theorem 1(b)–(c)) shows that the frontier is minimal over all safe local rules. Algorithm A's Layer 1 is the forward dominance of Dumas et al. (1991); Section 2.6 discusses rollback pruning (Lozano et al., 2016).

2.6 Pruning partial routes by cost

The frontier is computed from completed routes. A natural idea is to move part of the test into the enumeration, as rollback pruning does for the elementary shortest path problem (Lozano et al., 2016): compare a partial route (a label) L that has picked up an order x with the same partial route without the stops of x , and discard L and all its extensions if the extra distance and working time caused by x , priced at $\underline{\theta}$, already reach q_x . By Theorem 1(c) no rule of this kind can shrink the frontier; what is at stake is only the running time of Algorithm A.

Remark 10 (Rollback pruning needs a correction). Rollback pruning relies on the dominance relation of Feillet et al. (2004), in which time is a resource and arriving earlier never costs anything. Here working time is priced and waiting is allowed, so a time saving measured on the partial route need not survive to the end of the route: a route that arrives earlier at a stop that has not opened yet simply waits longer. Crediting the whole time saving is therefore unsafe (Example 2). The rule becomes safe with two corrections: a *junction term*, which measures the saving against the point where the continuation of L starts (otherwise the rule credits a detour that lies on the way to the next stop), and an *absorption term*, which subtracts the largest waiting that a stop still reachable by a continuation could impose. An earlier draft of this note proved that the corrected rule leaves every frontier unchanged, and a completion-level audit confirmed it: no violation among about 5.65 million completions of discarded labels, while removing either correction changed the frontier of two to four gigworkers.

Example 2 (Crediting the whole time saving is unsafe). On a synthetic instance (gigworker g_0 , $B = 3$) consider the label $L = (p_{o_2}, d_{o_2})$ and the order o_2 with $q_{o_2} = 17.13$. Without the absorption term the rule credits a time saving of 42.9 minutes and a cost saving of $17.72 \geq q_{o_2}$, so the label would be discarded. In the feasible completion $(p_{o_2}, d_{o_2}, p_{o_0}, d_{o_0})$, however, the pickup of o_0 opens late: without o_2 the driver would arrive early at p_{o_0} and wait. The true time saving is only 7.8 minutes and the true cost saving at $\underline{\theta}$ is $7.24 < 17.13$, so serving o_2 within this route is cheaper than delegating it. With the absorption term the label is kept.

Why the rule is not part of Algorithm A. Because the corrected rule must hold for every continuation, it can fire only once the detour caused by x is known and can no longer be absorbed by later waiting, that is, late. In the experiments 95–99% of the labels it removed already held B orders, so each removal saved only the remaining deliveries, whereas the test, which maintains a shortcut for every picked order, runs on every label that survives dominance. Added to the tiered version of Algorithm A it removed 14–32% of the extension attempts but made the run slower (0.60 – $0.80\times$); added to the final version ([Procedure 2](#)) it ran at 0.78 – $1.32\times$ (median $0.92\times$) over 41 instances, faster only where it removed at least about 40% of the remaining work, which cannot be known before the run. Many of its firings for occasional drivers were on prefixes that can no longer reach home in time, which the dead-end filter of [Proposition 3](#) removes far more cheaply. Evaluating the test earlier, before a pickup is visited, with a precomputed bound over all possible next stops, is safe but never fired on the synthetic instances: the median distance credit was 0.5 – 0.75 against FD prices of 15 – 21 , and the time credit was absorbed by waiting for later-released pickups. Comparing two generated labels with different sets of delivered orders (FD-dominance) can be made safe in the same way, and did not speed up Algorithm A either (0.81 – $1.01\times$). Details are in `EXPERIMENT_REPORT.md` and `EXPERIMENT_REPORT_C8.md`.

2.7 The final version of Algorithm A

The production code of Algorithm A processes labels in rounds of equal $|IV| + |C|$ and, inside a round, in repeated closures over deliveries; Layer 1 compares only labels created in the same round and the same closure step. Two labels with the same key (v, IV, C) can therefore be created in different batches and never be compared, and the dominated one is expanded (7.9–22.1% of the extension work on the five label instances, `EXPERIMENT_REPORT.md`). The final version processes labels in *tiers* of equal event count $m = |IV| + 2|C|$: every extension adds exactly one event, two labels with the same key have the same m , so every pair of same-key labels meets in one Layer-1 batch. Each label is filtered in exactly one tier and its extensions in later tiers, so [Lemma P3](#) holds verbatim. Moreover, all routes with k orders are complete once tier $2k$ has been processed, which allows the frontier to be computed inside the loop.

Proposition 2 (The frontier can be computed inside the loop). *Let R_i satisfy [Assumption A1](#) and $\theta < \bar{\theta}$, and let $R_i(S)$ be its routes with bundle S . Examine the bundles in order of nondecreasing size and accept a route $r \in R_i(S)$ if and only if $\mu'_r(b) > 0$ for some $b \in \Theta$, where μ'_r is the margin of [Definition 2](#) computed over the restricted set of substitutes*

$$D'(r) = \{\mathbf{0}_i\} \cup \bigcup_{S' \subsetneq S} \mathcal{K}'(S') \cup (R_i(S) \setminus \{r\}),$$

and $\mathcal{K}'(S')$ is the set of routes already accepted for the bundle S' . Then the accepted routes are exactly $\mathcal{K}_i^*$.

Proof. By induction on $|S|$; assume $\mathcal{K}'(S') = \mathcal{K}_i^* \cap R_i(S')$ for every bundle with fewer orders than S , and let $r \in R_i(S)$. Since $D'(r) \subseteq D(r)$, the restricted envelope is no lower than E_r , so $\mu'_r \geq \mu_r$; if $r \in \mathcal{K}_i^*$, then $\mu'_r(b) > 0$ for some b and r is accepted. If $r \notin \mathcal{K}_i^*$, fix $b \in \Theta$. [Lemma 1](#) gives $u \in \mathcal{K}_i^* \cup \{\mathbf{0}_i\}$, $u \neq r$, with $S_u \subseteq S$ and $c_u(b) + q(S \setminus S_u) \leq c_r(b)$. Either $u = \mathbf{0}_i$; or $S_u \subsetneq S$, and then $u \in \mathcal{K}_i^* \cap R_i(S_u) = \mathcal{K}'(S_u)$ by induction; or $S_u = S$ and $u \in R_i(S) \setminus \{r\}$. In every case $u \in D'(r)$, so $\mu'_r(b) \leq 0$. As b was arbitrary, r is rejected. $\square$

In `EXPERIMENT_REPORT.md` the tiered version with the in-loop frontier is called C5; its in-loop frontier was validated by tests only, and [Proposition 2](#) is the missing proof.

Removing dead ends. The production code checks the detour budget of an occasional driver only when the home leg is added, and checks a delivery’s closing time only when that delivery is added. Labels that can no longer be completed are therefore extended to the end; on the five label instances, occasional drivers account for 50–62% of all extension attempts, while their frontier holds 6–7 routes.

Proposition 3 (Removing dead ends). *Call a label $L = (v, IV, C, t, K, W)$ dead if $t + \tau(v, d_j) > l_{d_j}$ for some $j \in IV$, or if the driver is an occasional driver with home h and home deadline l_h , and*

$$\max\left\{t + \tau(v, h), \max_{j \in IV}(\max\{t + \tau(v, d_j), e_{d_j}\} + s_{d_j} + \tau(d_j, h))\right\} > l_h.$$

Discarding every dead label when it is created leaves R_i , and hence $\mathcal{K}_i^$, unchanged.*

Proof. Call a label *useless* if it has no feasible completion. (i) A dead label is useless: any completion reaches d_j no earlier than $t + \tau(v, d_j)$ and, for an occasional driver, reaches home no earlier than the left-hand side above (triangle inequality, nonnegative waiting and service). (ii) Extensions of a useless label are useless, since their completions are completions of the label. (iii) A label dominated by a useless label is useless: if L' , with the same key as L and no earlier clock, had a feasible completion σ , then $L \cdot \sigma$ would be feasible by (4). By (ii), the filter removes only useless labels and prevents only useless labels from being created. By (iii), a label that is not useless can be dominated only by labels that are not useless; by induction over the batches, these are present with and without the filter, so exactly the same labels that are not useless survive Layer 1. Every completed route comes from such labels, so the completed routes, the Pareto fronts of Layer 2 and therefore R_i are unchanged. $\square$

Tiers, the dead-end filter and the in-loop frontier together give Algorithm A in its final form (Procedure 2). It never stores R_i and outputs $\mathcal{K}_i^*$ directly; Section 2.8 reports its running time.

2.8 Computational evidence

Synthetic audit of the theory. The statements of Section 1 were checked against a reference implementation that shares no code with the production implementation: it enumerates every precedence-feasible sequence of every bundle, computes the frontier analytically and solves every winner-determination problem exactly (HiGHS, zero gap). The instances have $n = 8$ orders in a 10×10 km square, two gigworkers and two occasional drivers, $B = 3$, $\Theta = [18, 25]$, FD tariff $8 + 3 \max\{0, d - 2\}$, $\kappa = 0.5$, speed 20 km/h and two minutes of service per stop; nine seeds. These checks test the mathematics; they are not thesis results.

Experimental instances: safety and pool reduction. On 600 instances of the main grid (alignment 0.50 and 0.90, all sizes and supply mixes), every base and removal problem was solved on the full pool and on the frontier. Across 1,924 solves per variant, the largest deviation in any optimal value or payment was 1.1×10^{-13} . Table 3 reports the effect on the payment step.

The frontier as a pre-auction certificate. An empty frontier certifies, before any bid is collected, that a driver has no route worth considering at any admissible report. Table 4 separates this from the trivial case of a driver without any feasible route. Consistently with Remark 5, at alignment 0.50 and $n = 15$ the certified share among drivers with a feasible route falls from 92.3% to 59.2% and 23.1% as the FD tariff multiplier rises from 0.75 to 1.00 and 1.25, while the share without a feasible route stays at 43.7%.

Procedure 2 GENERATEROUTES(i): Algorithm A for one driver (final version: event tiers, dead-end filter, frontier inside the loop)

Input: public data of driver i ; orders O ; cap B ; FD prices q ; $\Theta = [\underline{\theta}, \bar{\theta}]$

Output: local frontier $\mathcal{K}_i^*$

```

1:  $\mathcal{L} \leftarrow \{(v_0, \emptyset, \emptyset, t_0, 0, 0)\}$ ;  $m \leftarrow 0$ ;  $\mathcal{C}[S] \leftarrow \emptyset$  and  $\mathcal{K}'(S) \leftarrow \emptyset$  for every bundle  $S$ 
2: while  $\mathcal{L} \neq \emptyset$  do ▷  $\mathcal{L}$  = labels with  $m = |IV| + 2|C|$  events
3:    $\mathcal{L} \leftarrow \text{FILTERDOMINATED}(\mathcal{L})$  ▷ Layer 1: one batch per tier, key  $(v, IV, C)$ 
4:    $\mathcal{L}' \leftarrow \emptyset$ 
5:   for each label  $\ell \in \mathcal{L}$  do
6:     if  $IV(\ell) = \emptyset$  and  $C(\ell) \neq \emptyset$  then
7:       add  $\ell$  (gigworker) or TRYHOME( $\ell$ ) (occasional driver, if feasible) to  $\mathcal{C}[C(\ell)]$ 
8:     end if
9:     add every feasible TRYDELIVERY( $\ell, j$ ),  $j \in IV(\ell)$ , that is not dead to  $\mathcal{L}'$  ▷ Proposition 3
10:    if  $|IV(\ell)| + |C(\ell)| < B$  then
11:      add every feasible TRYPICKUP( $\ell, j$ ),  $j \notin IV(\ell) \cup C(\ell)$ , that is not dead to  $\mathcal{L}'$ 
12:    end if
13:  end for
14:  if  $m \geq 2$  is even then ▷ every route with  $k = m/2$  orders is now complete
15:    for each bundle  $S$  with  $|S| = m/2$  and  $\mathcal{C}[S] \neq \emptyset$  do
16:       $F \leftarrow \text{PARETOFRONT}(\{(K_\ell, W_\ell) : \ell \in \mathcal{C}[S]\})$  ▷ Layer 2
17:       $\mathcal{K}'(S) \leftarrow \{r \in F : \mu'_r(b) > 0 \text{ for some } b \in \Theta\}$  ▷ Proposition 2, test as in Procedure 1
18:    end for
19:  end if
20:   $\mathcal{L} \leftarrow \mathcal{L}'$ ;  $m \leftarrow m + 1$ 
21: end while
22: return  $\mathcal{K}_i^* = \bigcup_S \mathcal{K}'(S)$ 

```

Table 2: Synthetic audit of Theorems 1 and 2.

Check	Result
Violations of Assumption A2 (all subsets, three reports)	0
Routes after Pareto filtering / kept by Π^*	2,985 / 1,157 (38.8%)
gigworkers / occasional drivers	1,123 of 2,908 / 34 of 77
Theorem 1(a): max. $ V(R^{\Pi^*}) - V(R) $ over base and removal solves	5.7×10^{-14}
Corollary 1: max. payment difference (1,737 payments)	5.7×10^{-14}
Theorem 1(b): frontier routes made indispensable by the construction	1,157 / 1,157
smallest increase of the optimal value when the route is removed	3.4×10^{-4}
Remark 2: indispensable against geometric competitors (four seeds)	556 / 563
Frontier routes deleted by a per-driver hull across bundles	1,144 / 1,157
Routes optimal for some of 400 report profiles (three seeds)	1.9–3.8% of pool, all in $\mathcal{K}^*$
Theorem 2, $n = 3, 5, 7, 9$: $ \mathcal{K}_g^* = \sum_k \binom{n}{k}$ and g never optimal	confirmed
Mutation: FD term removed from E_r	detected
Independent grid-based margin (5,000 points) vs. breakpoint method	42 / 42 pools agree

Table 3: Payment computation on the full pool (naive) and on the frontier (600 instances, $B = 3$). Times in seconds, medians.

Alignment	n	Pool removed	t_C naive	t_C frontier	Speed-up [IQR]	Incl. construction
0.50	10	100.0%	0.039	0.003	11.20 [10.11, 13.14]	1.13
0.50	15	100.0%	0.086	0.005	17.98 [14.62, 22.05]	0.63
0.50	20	100.0%	0.186	0.006	30.40 [21.92, 41.85]	0.60
0.90	10	75.2%	0.134	0.060	2.34 [1.69, 3.04]	1.44
0.90	15	76.6%	0.248	0.144	1.81 [1.56, 2.57]	1.05
0.90	20	77.2%	0.655	0.259	2.72 [2.11, 3.37]	1.24
Pooled over 600 instances					3.01	1.06

Table 4: Drivers without a feasible route, and drivers certified in advance as dominated by FD ($R_i \neq \emptyset$, $\mathcal{K}_i^* = \emptyset$), main grid (7,500 drivers).

Alignment	0.10	0.30	0.50
No feasible route (% of drivers)	14.8	12.3	43.1
Certified FD-dominated (% of drivers)	20.4	19.5	34.3
Certified, among drivers with a feasible route (%)	23.9	22.3	60.2
Instances in which no auction is needed (%)	8.3	7.7	53.0

The final version of Algorithm A. Table 5 reports the time to the frontier of the production code (C1), of the tiered version with the in-loop frontier (C5) and of the final version, which adds the dead-end filter (C5h, Procedure 2). The filter removes 52–76% of the extension attempts of C5, and 99% of those of occasional drivers; the frontier is unchanged on every instance, as Proposition 3 guarantees, and an independent audit found no removed label with a feasible completion (12,182 labels checked). The final version is $1.9\text{--}3.9\times$ faster than the production code and $1.6\text{--}2.9\times$ faster than C5; on the main-grid sample it is $2.1\times$ (alignment 0.90) and $2.3\times$ (alignment 0.50) faster than the production code. A stronger, still exact, dead-end test, which asks whether the orders on board can be delivered in *some* order before their deadlines, removed at most 0.3% of the remaining extension attempts and made the run slower ($0.70\text{--}0.84\times$ of C5h): almost every label that cannot be completed is already caught by Proposition 3. What remains is work on labels that *can* be completed; on the label instances, 91–99% of the gigworkers’ completed routes are not in the frontier. Removing them earlier requires a cost argument, and Section 2.6 explains why the one we studied does not pay off.

Table 5: Time to the frontier (seconds, median of ten timed runs) of the production Algorithm A (C1), the tiered version with the frontier inside the loop (C5) and the final version with the dead-end filter (C5h); speed-ups are medians of paired ratios. Pure Python 3.13, one core of an Intel Xeon at 2.1 GHz.

Instance	B	C1	C5	C5h	C5h vs C1	C5h vs C5
$n = 12$, seed 42	3	0.143	0.121	0.052	2.73	2.36
$n = 10$, seed 1	3	0.131	0.108	0.068	1.92	1.57
$n = 15$, seed 7	3	0.442	0.361	0.177	2.46	2.04
$n = 12$, seed 123	3	0.274	0.221	0.137	1.97	1.60
$n = 10$, seed 999	3	0.089	0.075	0.039	2.21	1.85
$n = 12$, seed 42	4	1.391	1.025	0.349	3.90	2.86
$n = 10$, seed 1	4	1.437	1.000	0.611	2.33	1.63
$n = 15$, seed 7	4	7.455	5.521	2.503	3.00	2.25
$n = 12$, seed 123	4	3.294	2.211	1.342	2.43	1.64
$n = 10$, seed 999	4	0.758	0.531	0.275	2.77	1.90

2.9 Items to verify before submission

- Read [Sol \(1994\)](#) and [Vanderbeck \(1994\)](#) directly; the statements in [Remark 9](#) are taken from [Lübbecke and Desrosiers \(2005\)](#).
- Confirm that the route deleted in [Example 2](#) belongs to the frontier of g_0 , and the exact criterion behind 556/563 in [Remark 2](#).
- State the numerical tolerances used by [Procedures 1](#) and [2](#).
- Bibliographic details of all references below.

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